

Classwork 08//Diffusion Process and Gaussian Distribution

Student Name: _____ **Points:** 30

A Number: _____ **Due Date:** April 07, 2026

Consider a random variable

$$\varepsilon \sim \mathcal{N}(0, \mathbf{I})$$

Imagine that $\mathbf{x}_0$ is the initial data distribution (the original data distribution). Then, in the forward diffusion process, we have

$$\mathbf{x}_{t+1} = \sqrt{1 - \beta_t} \mathbf{x}_t + \sqrt{\beta_t} \varepsilon$$

Here, the recursive relationship collapses into a single step relative to $\mathbf{x}_0$:

$$\mathbf{x}_{t+1} = \sqrt{\bar{\alpha}_{t+1}} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t+1}} \varepsilon$$

where we have the following relationship:

$$\alpha_i = 1 - \beta_i, \text{ and } \bar{\alpha}_{t+1} = \prod_{i=0}^t \alpha_i$$

Suppose somebody told you that initial data $\mathbf{x}_0$ has distribution $\mathcal{N}(m, s)$ where s is the variance and m is the mean.

1. Derive the mean $\mathbb{E}[\mathbf{x}_{t+1}]$ and the variance $\mathbb{V}(\mathbf{x}_{t+1})$ in the form of β_{t+1} , m and s .
2. Also provide the expression when $\beta_{t+1} = 0.1$ for all t .
3. Derive $q(\mathbf{x}_1 | \mathbf{x}_0)$.
4. In Diffusion process, in every step, we make sure that variance of the noisy data at time t doesn't explode. This is called variance preserving method. How does it guide us in choose what β_t we can use? How is it related to stochastic differential equation (SDE)?

In doing above derivation, please state what properties of expectations, and Gaussian distributions you are using.

Solution

1. Expectations:

$$\begin{aligned}
 \mathbf{x}_{t+1} &= \sqrt{\bar{\alpha}_{t+1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t+1}}\varepsilon \\
 \mathbb{E}[\mathbf{x}_{t+1}] &= \mathbb{E}[\sqrt{\bar{\alpha}_{t+1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t+1}}\varepsilon] \\
 &= \sqrt{\bar{\alpha}_{t+1}}\mathbb{E}[\mathbf{x}_0] + \sqrt{1 - \bar{\alpha}_{t+1}}\mathbb{E}[\varepsilon] \quad \text{Property of Expectations} \\
 &= \sqrt{\bar{\alpha}_{t+1}}m + \sqrt{1 - \bar{\alpha}_{t+1}} \cdot 0 \quad \text{(Given)} \quad \varepsilon \sim \mathcal{N}(0, \mathbf{I}) \\
 &= \sqrt{\bar{\alpha}_{t+1}}m
 \end{aligned}$$

Variance:

$$\begin{aligned}
 \mathbf{x}_{t+1} &= \sqrt{\bar{\alpha}_{t+1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t+1}}\varepsilon \\
 \mathbb{V}[\mathbf{x}_{t+1}] &= \mathbb{V}[\sqrt{\bar{\alpha}_{t+1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t+1}}\varepsilon] \\
 \mathbb{V}[\mathbf{x}_{t+1}] &= (\sqrt{\bar{\alpha}_{t+1}})^2\mathbb{V}[\mathbf{x}_0] + (\sqrt{1 - \bar{\alpha}_{t+1}})^2\mathbb{V}[\varepsilon] \\
 &= \bar{\alpha}_{t+1}\mathbb{V}[\mathbf{x}_0] + (1 - \bar{\alpha}_{t+1})\mathbb{V}[\varepsilon] \\
 &= \bar{\alpha}_{t+1}s + (1 - \bar{\alpha}_{t+1}) \cdot 1 \quad \text{(Given)} \quad \varepsilon \sim \mathcal{N}(0, \mathbf{I})
 \end{aligned}$$

2.

$$\begin{aligned}
 \beta_{t+1} &= 0.1 \forall t \\
 \alpha_t &= 1 - \beta_t = 0.9 \\
 \bar{\alpha}_{t+1} &= \prod_{i=0}^t \alpha_i = 0.9^{t+1}
 \end{aligned}$$

Hence,

$$\begin{aligned}
 \mathbb{E}[\mathbf{x}_{t+1}] &= 0.9^{\frac{t+1}{2}}m \\
 \mathbb{V}[\mathbf{x}_{t+1}] &= 0.9^{t+1}s + (1 - 0.9^{t+1})
 \end{aligned}$$

3.

$$q(\mathbf{x}_1 | \mathbf{x}_0) = ?$$

Setting $t = 0$,

$$\mathbf{x}_1 = \sqrt{1 - \beta_0}\mathbf{x}_0 + \sqrt{\beta_0}\varepsilon$$

When we condition on a specific value of $\mathbf{x}_0$, the term $\mathbf{x}_0$ is treated as a constant. The only source of randomness in the equation is the Gaussian noise ε . We can find the parameters of $\mathbf{x}_1$ using the properties of linear transformations of Gaussian variables:

$$\begin{aligned}\mathbb{E}[\mathbf{x}_1|\mathbf{x}_0] &= \mathbb{E}[\sqrt{1-\beta_0}\mathbf{x}_0 + \sqrt{\beta_0}\varepsilon] \\ \mathbb{E}[\mathbf{x}_1|\mathbf{x}_0] &= \sqrt{1-\beta_0}\mathbf{x}_0 + \sqrt{\beta_0}\mathbb{E}[\varepsilon]\end{aligned}$$

Since $\mathbb{E}[\varepsilon] = 0$, the mean is:

$$\mathbb{E}[\mathbf{x}_1|\mathbf{x}_0] = \sqrt{1-\beta_0}\mathbf{x}_0$$

For Variance:

$$\mathbb{V}(\mathbf{x}_1|\mathbf{x}_0) = \mathbb{V}(\sqrt{1-\beta_0}\mathbf{x}_0 + \sqrt{\beta_0}\varepsilon)$$

Because $\mathbf{x}_0$ is constant under the conditioning, its variance is 0.

$$\mathbb{V}(\mathbf{x}_1|\mathbf{x}_0) = (\sqrt{\beta_0})^2\mathbb{V}(\varepsilon)$$

Since $\mathbb{V}(\varepsilon) = \mathbf{I}$, the variance is:

$$\mathbb{V}(\mathbf{x}_1|\mathbf{x}_0) = \beta_0\mathbf{I} = \beta_0$$

(if we consider one dimension).

Thus

$$q(\mathbf{x}_1|\mathbf{x}_0) \sim \mathcal{N}(\mathbf{x}_1; \sqrt{1-\beta_0}\mathbf{x}_0, \beta_0\mathbf{I})$$

4. The Variance Preserving (VP) method is a specific design for the forward diffusion process that ensures the latent variables $\mathbf{x}_t$ maintain a constant scale throughout the transition from data to noise.

In the forward process, the transition is:

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\varepsilon$$

This is Variance Preserving because if we assume $\mathbf{x}_{t-1}$ has unit variance ($\mathbb{V}(\mathbf{x}_{t-1}) = 1$), then the variance of $\mathbf{x}_t$ remains 1:

$$\mathbb{V}(\mathbf{x}_t) = (\sqrt{1-\beta_t})^2 \cdot 1 + (\sqrt{\beta_t})^2 \cdot 1 = 1 - \beta_t + \beta_t = 1$$

Contrast this with a simple random walk (like Variance Exploding methods), where you simply add noise: $\mathbf{x}_t = \mathbf{x}_{t-1} + \sigma_t \epsilon$. In that case, the variance would grow to infinity as t increases.

But the VP formulation acts as a guardrail for the generative model.

Because the total variance is kept near 1, the model doesn't have to deal with exploding numerical values. The data is effectively squashed toward the origin as noise is added, ensuring $\mathbf{x}_T$ always lands within the neat confines of a standard normal distribution $\mathcal{N}(0, \mathbf{I})$.

VP diffusion can be viewed as a discretization of the following SDE:

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x} dt + \sqrt{\beta(t)} d\mathbf{w}$$

Here, the first term $(-\frac{1}{2}\beta(t)\mathbf{x})$ is a drift term that constantly pulls the distribution back toward the origin, preventing it from wandering off, while the second term adds the diffusion.